

The empirical recurrence for OEIS A252905, recovered from its tail

Adrian Perez Fontelles
Independent researcher

8 September 2026

Abstract

OEIS A252905 records “Empirical recurrence of order 63” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 135$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 2$, so the recurrence is proved for every $n \geq 65$. Its last coefficient is $c_{63} = -84934656 \neq 0$, so no further tail satisfies anything shorter; any order-63 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture that is not written down

OEIS A252905 is “Number of $(n+2) \times (1+2) \dots 3$ arrays with every consecutive three elements in every row and column having exactly 2 distinct values, in every diagonal 1 or 2 distinct values, in every antidiagonal 2 or 3 distinct values, and new values 0 upwards introduced in row major order.”. It has offset 1 and begins

238, 1125, 5567, 28642, 148002, 771768, 4034881, 21118867, ...

The entry states:

Empirical recurrence of order 63 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 13:44:52, revision 7) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry's shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 135$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 135$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 63: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 2$ is the first whose minimal order is exactly the stated 63.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 270$ exact terms from $n = 2$ on, it returns order 63 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{63} c_i a(n-i),$$

c_1	5	c_{17}	35781820	c_{33}	−29429863100	c_{49}	1420445097704
c_2	19	c_{18}	−10059864	c_{34}	101879826028	c_{50}	247401907816
c_3	−32	c_{19}	−103199034	c_{35}	141473942184	c_{51}	−901211721328
c_4	−401	c_{20}	−23992731	c_{36}	−85687722460	c_{52}	−474364331096
c_5	−641	c_{21}	378704591	c_{37}	−294329300580	c_{53}	551977177552
c_6	4381	c_{22}	177214518	c_{38}	−3763500890	c_{54}	254052723456
c_7	12266	c_{23}	−1100908089	c_{39}	418729143162	c_{55}	−343968674272
c_8	−8203	c_{24}	−1235752573	c_{40}	282799791242	c_{56}	−23823127424
c_9	−112475	c_{25}	1966212676	c_{41}	−533609432732	c_{57}	145593693504
c_{10}	−163052	c_{26}	4502388529	c_{42}	−825881041356	c_{58}	−24662965632
c_{11}	482779	c_{27}	−2759495509	c_{43}	306339107260	c_{59}	−30988920320
c_{12}	1367338	c_{28}	−9982532526	c_{44}	1306639715336	c_{60}	6827814912
c_{13}	273307	c_{29}	4748954645	c_{45}	597116225884	c_{61}	2611666944
c_{14}	−5743409	c_{30}	24592089400	c_{46}	−1203523828056	c_{62}	−360972288
c_{15}	−9637366	c_{31}	−3498026801	c_{47}	−1450322249348	c_{63}	−84934656
c_{16}	13859618	c_{32}	−64559779143	c_{48}	488996818512		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 65$, and it is the recurrence OEIS A252905 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 2$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{63} - c_1 x^{62} - \dots - c_{63}$. Its constant term is $-c_{63} = 84934656$, which is not zero, so $q_{\min}(0) \neq 0$

and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 63 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 63, so $\deg q = 63$, and it is monic. It holds from some index t on. From $\max(t, 2)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 63, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 19 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 63, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -84934656 .

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-63 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A252905.
- [2] J. L. Massey, Shift-register synthesis and BCH decoding, *IEEE Trans. Inform. Theory* **15** (1969), 122–127.
- [3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)
- [4] M. Kauers and P. Paule, *The Concrete Tetrahedron*, Springer, 2011. (C-finite sequences and their closure properties.)
- [5] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.